

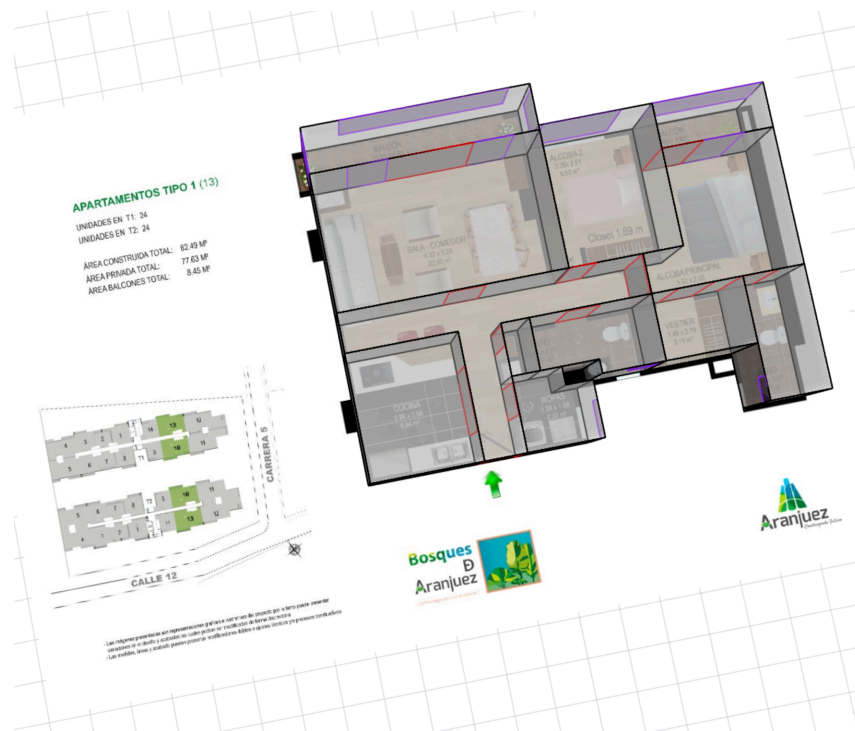
Transformation of an architectural model into an interpretable spatial model using TopologicPy

Assignment 01

Arq. David Agudelo

May 2026

Aranjuez is an apartment building with two towers of eight floors, one basement, and one access level facing the street. This project focuses on the analysis of a private unit to identify the Housing as a network of connections, identifying central nodes, studying direct relationships and inferring patterns of circulation or spatial dependence.

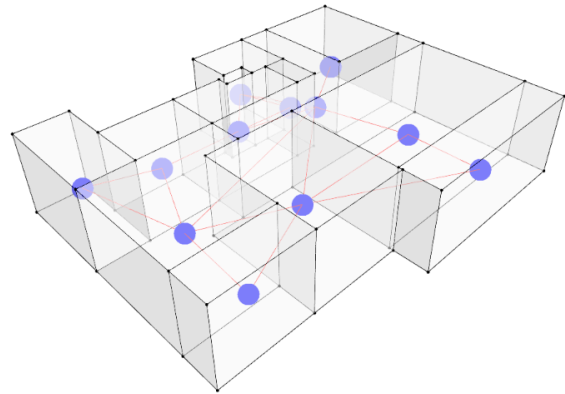
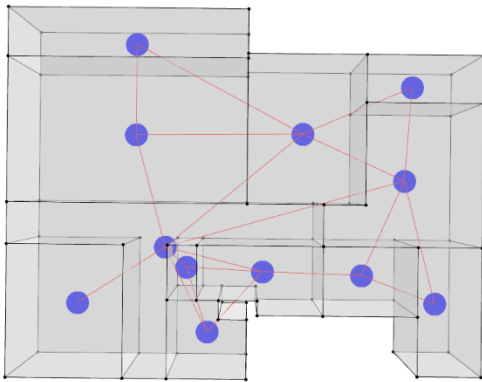


Transformation of an architectural model into an interpretable spatial model using TopologicPy

Assignment 01

Arq. David Agudelo

May 2026



The model ceases to be merely a composition of surfaces or volumes and becomes a system of interconnected spaces, where it is possible to study relationships of proximity, adjacency, and access.

Each cell represents a recognizable architectural space within the dwelling. Subsequently, these spaces are assigned attributes using semantic dictionaries, so that each can be identified by name, position, and associated properties. This incorporation of non-geometric information is key because it allows the model to be interpreted not only by its form but also by its function and role within the whole.

From the files exported by the notebook, the following global summary of the analyzed model is obtained:

| Indicator | Value |

| Identified spaces | 12 |

| Total openings | 22 |

| Graph vertices | 35 |

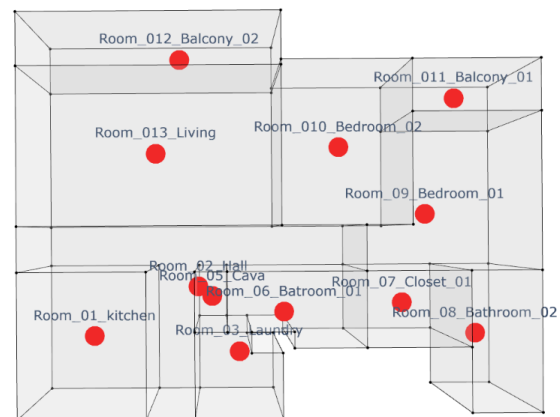
Graph edges | 37 |

Spatial Relationships Through Openings

The `apertures\_summary.csv` file allows us to recognize how different spaces are connected through openings. Among the most representative relationships in the model are the following:

1	id	type	name	area	width	height	center_x	center_y	center_z	connected_room_count	connected_rooms
2	A001	aperture	Aperture_001	1.946	2.224	0.875	8.711	14.081	1.112	2	Room_02_Hall Room_013_Living
3	A002	aperture	Aperture_002	3.411	2.231	1.529	8.274	11.567	1.115	2	Room_02_Hall Room_01_kitchen
4	A003	aperture	Aperture_003	2.372	2.4	0.989	14.278	16.418	1.2	2	Room_09_Bedroom_01 Room_011_Balcony_01
5	A004	aperture	Aperture_004	4.882	2.4	2.034	8.5	17.399	1.2	2	Room_013_Living Room_012_Balcony_02
6	A005	aperture	Aperture_005	2.057	2.328	0.883	15.427	13.127	1.164	2	Room_09_Bedroom_01 Room_08_Bathroom_02
7	A006	aperture	Aperture_006	2.051	2.4	0.855	13.92	13.127	1.2	2	Room_09_Bedroom_01 Room_07_Closet_01
8	A007	aperture	Aperture_007	1.654	2.134	0.775	11.285	13.127	1.067	2	Room_02_Hall Room_06_Bathroom_01
9	A008	aperture	Aperture_008	2.01	2.249	0.894	12.83	13.634	1.124	2	Room_02_Hall Room_09_Bedroom_01
10	A009	aperture	Aperture_009	1.488	2.4	0.62	11.433	14.081	1.2	2	Room_02_Hall Room_010_Bedroom_02
11	A010	aperture	Aperture_010	1.695	2.231	0.76	9.277	11.361	1.115	2	Room_03_Laundry Room_02_Hall
12	A011	aperture	Aperture_011	1.097	2.326	0.472	9.277	12.841	1.163	2	Room_05_Cava Room_02_Hall
13	A012	aperture	Window_07	1.241	1.398	0.888	11.082	17.953	1.701	1	Room_012_Balcony_02
14	A013	aperture	Window_07	1.423	1.398	1.018	5.614	17.908	1.701	1	Room_012_Balcony_02
15	A014	aperture	Window_07	5.673	4.058	1.398	8.553	18.519	1.701	1	Room_012_Balcony_02
16	A015	aperture	Window_07	3.708	2.652	1.398	15.11	17.509	1.701	1	Room_011_Balcony_01
17	A016	aperture	Window_07	0.371	0.626	0.593	14.985	10.481	2.087	1	Room_08_Bathroom_02
18	A017	aperture	Window_07	0.517	0.842	0.614	12.389	11.542	1.932	2	Room_02_Hall Room_06_Bathroom_01
19	A018	aperture	Window_07	0.58	0.898	0.646	11.093	10.535	2.064	2	Room_03_Laundry Room_02_Hall
20	A019	aperture	Window_07	2.301	2.4	0.959	7.004	17.399	1.2	2	Room_013_Living Room_012_Balcony_02
21	A020	aperture	Window_07	2.31	2.4	0.963	9.999	17.399	1.2	2	Room_013_Living Room_012_Balcony_02
22	A021	aperture	Window_07	3.784	2.025	1.869	12.126	17.509	1.388	1	Room_010_Bedroom_02
23	A022	aperture	Window_07	2.362	2.4	0.984	15.265	16.418	1.2	2	Room_09_Bedroom_01 Room_011_Balcony_01

- Hall - Living
- Hall - Kitchen
- Bedroom_01 - Balcony_01
- Living - Balcony_02
- Bedroom_01 - Bathroom_02
- Bedroom_01 - Closet_01
- Hall - Bathroom_01
- Hall - Bedroom_01
- Hall - Bedroom_02
- Laundry - Hall
- Cava - Hall



These relationships allow us to identify that the hall fulfills a central distribution function within the dwelling. The repeated presence of connections from this space to main and service rooms shows that it acts as an articulating node of the spatial system. Consequently, the model represents not only isolated rooms, but a network organized around a dominant transitional space.

Spatial Relationships Through Openings

The spatial model can reveal functional hierarchies. Spaces with greater connectivity tend to correspond to distribution, circulation, or articulation zones, while those with less connectivity tend to correspond to more private or specialized uses.

